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Started	Monday, 19 October 2026, 11:52 AM
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Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Sorting the entire list sequentially
b. Combining sorted sublists
c. Dividing the input into smaller subproblems
d. Comparing each element with the others

The correct answer is: Dividing the input into smaller subproblems

Question **2**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

a. Quick Sort
b. Bubble Sort
c. Merge Sort
d. Insertion Sort

The correct answer is: Bubble Sort

Question **3**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

a. Bubble Sort is difficult to implement
b. Bubble Sort is less efficient for large lists
c. Bubble Sort cannot handle duplicate elements
d. Bubble Sort does not guarantee sorted order

The correct answer is: Bubble Sort is less efficient for large lists

Question **4**

Complete

Mark 0.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 40,70,80,90,120,150,200,300,400,470
- b. 80,150,200,470,40,90,300,400,70,120
- c. 40,80,90,150,200,300,400,470,70,120
- d. 200,470,80,150,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **5**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

- a. Sorting
- b. Rearranging
- c. Merging
- d. Searching

The correct answer is: Sorting

Question **6**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

- a. Merge Sort
- b. Binary Search
- c. Bubble Sort
- d. Linear Search

The correct answer is: Bubble Sort

Question **7**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

a. It increases the number of comparisons needed
b. It makes the search operation slower
c. It has no effect on the search operation
d. It allows for faster searching

The correct answer is: It allows for faster searching

Question **8**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

a. They are split again into smaller sublists
b. They are discarded
c. They are combined to form a single sorted list
d. They are compared element by element

The correct answer is: They are combined to form a single sorted list

Question **9**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

a. Sorting makes it harder to search for items
b. Sorting helps in finding duplicates quickly
c. Sorting decreases the efficiency of selection operations
d. Sorting is rarely used in programming

The correct answer is: Sorting helps in finding duplicates quickly

Question **10**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

a. Deleting data from a list
b. Inserting data into a list
c. Searching for data in a list
d. Arranging data in a particular format

The correct answer is: Arranging data in a particular format

Question **11**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

a. Sorting data in a single pass
b. Rearranging data without sorting
c. Dividing the input into parts, solving each part, and combining the solutions
d. Sorting data sequentially

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **12**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It avoids the need for recursion
b. It simplifies the sorting process by using only one pass
c. It only works on small datasets
d. It allows for efficient parallel processing and sorting of data

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **13**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

a. Makes code execution slower
b. Makes data harder to manage
c. Decreases the efficiency of algorithms
d. Provides a basis for other algorithms to work efficiently

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **14**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

a. By comparing elements sequentially
b. By sorting data in a single pass
c. By using the bubble-up method
d. By breaking the input into smaller parts and merging them

The correct answer is: By breaking the input into smaller parts and merging them

Question **15**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

a. Searching
b. Selection
c. All of the above
d. Duplicates

The correct answer is: All of the above